

Notes: Da, Engelberg, and Gao (JF, 2011)

In Search of Attention

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1 Big picture

- **Question.** Can internet search activity provide a direct measure of investor attention, and does a rise in that attention create temporary stock price pressure that later reverses?
- **Setting.** The paper studies weekly Google Search Volume Index (SVI) data for Russell 3000 stocks from 2004 to 2008, combines them with news, returns, turnover, analyst coverage, and Dash-5 retail order-flow data, and then applies the same idea to U.S. IPOs from 2004 to 2007.
- **Core challenge.** Investor attention is usually unobserved. Standard proxies such as extreme returns, turnover, news coverage, or advertising are indirect and can move for reasons unrelated to attention itself.
- **Main contribution.** The paper proposes search intensity in Google as a revealed-attention measure. If an investor searches a ticker, that investor is almost surely paying attention to the stock. The paper then shows that this measure is distinct from existing proxies, is closely tied to retail investor trading, and has strong predictive power for short-run return pressure and long-run reversal.
- **Why it matters.** Limited-attention theories in finance need a measure of actual attention, not just equilibrium outcomes. This paper provides one and uses it to test the Barber–Odean attention-induced price pressure hypothesis.
- **Main empirical question.** When search-based attention rises, do retail investors buy more, push prices up in the short run, and generate later reversals?
- **Main empirical results.**
 - Search volume is positively correlated but only weakly with existing attention proxies such as turnover, extreme returns, and news.
 - In a VAR, SVI leads turnover, extreme returns, and chunky news, suggesting that it captures attention in a more timely way.
 - A 1% increase in SVI is associated with roughly a 0.103% increase in retail orders and a 0.131% increase in retail turnover in Dash-5 data.
 - The relation is much stronger at Madoff, a market center associated with less sophisticated retail order flow, than at NYSE/Archipelago.
 - A one-standard-deviation increase in ASVI predicts about 18.7 bps of abnormal return in the next week and an eventual reversal of about 28.9 bps over weeks 5 to 52.
 - These effects are stronger for smaller stocks and for stocks with more retail trading.
 - Around IPOs, search activity spikes sharply. IPOs with high pre-IPO abnormal search have higher first-day returns and worse long-run performance when first-day returns are also high.
- **Economic takeaway.** Attention is a scarce resource, retail investors chase attention-grabbing stocks, and search data reveal that these attention shocks create temporary demand pressure rather than permanent information effects.

2 Data and measurement (key objects)

2.1 Google Trends and search-based attention

- **Main source.** Google Trends reports a weekly Search Volume Index (SVI) for a search term. SVI is the number of searches for that term scaled by its own time-series average.
- **Why ticker searches?** The paper mostly uses stock tickers rather than company names. Ticker searches are less ambiguous: a search for “AAPL” is much more likely to reflect investor attention than a search for “Apple,” which could be about products or shopping.
- **Main stock sample.** The authors study all 3,606 firms ever included in the Russell 3000 between January 2004 and June 2008. This yields 834,627 firm-week observations, of which 468,413 have valid SVI values.
- **Large-stock focus.** Because the Russell 3000 contains large firms, the sample is less affected by bid-ask bounce and other microstructure issues than a very small-stock sample would be.
- **Name SVI.** For comparison, the paper also constructs search intensity using company names. This variable is much noisier because names can have multiple meanings and because users may search for the same firm in different ways.
- **Product search.** The paper also constructs **PSVI**, search volume for the firm’s main product, based on Nielsen advertising data and hand matching. This is used to distinguish retail attention from product-demand news.

2.2 The key abnormal-search variable

- The main variable is **abnormal SVI**, denoted ASVI, which measures unexpected attention relative to the stock’s own recent history.
- The benchmark level of attention is the **median** SVI over the previous 8 weeks. Using the median makes the measure robust to temporary spikes.
- This is important because raw SVI levels differ enormously across stocks; ASVI is the cross-sectionally comparable attention surprise.

2.3 Other attention, sentiment, and market variables

- **CRSP-based variables.** Returns, characteristic-adjusted abnormal returns, turnover, and abnormal turnover.
- **News variables.** The Dow Jones archive provides article counts. The paper distinguishes simple article counts from **Chunky News**, stories with multiple codes or multiple newswire messages, meant to isolate more important news.
- **Sentiment variables.** Negative-word shares are built from media text using the Harvard IV-4 and Loughran–McDonald dictionaries.
- **Firm controls.** The regressions also use market capitalization, analyst coverage from I/B/E/S, and advertising-expense-to-sales from COMPUSTAT.

2.4 Retail trading: Dash-5 data

- To identify who is searching, the paper uses **SEC Rule 11Ac1-5 (Dash-5)** monthly reports.
- Dash-5 reports mostly cover **retail orders**, because institutional not-held orders are excluded and very large orders above 10,000 shares are not reported.
- The paper studies monthly changes in
 - the number of retail orders,
 - retail turnover,
 - and the share of overall trading volume accounted for by Dash-5, **Percent Dash-5 Volume**.
- It also separates market centers. Orders routed to **Madoff** are treated as reflecting less sophisticated retail investors, whereas **NYSE/Archipelago** is treated as reflecting more sophisticated investor flow.

2.5 IPO data

- The IPO sample comes from the SDC new issues database. After standard filters, the paper identifies 571 common-share IPOs on NYSE, Amex, and NASDAQ from 2004 to 2007.
- For IPOs, the paper uses **company-name searches** rather than ticker searches because pre-IPO tickers are not widely available.
- Only 185 IPOs have valid SVI values because of naming ambiguity and Google truncation when search activity is too low.
- Additional IPO controls include
 - **Media**: pre-IPO Factiva news coverage,
 - **Price Revision**: offer price relative to the filing range median,
 - **DSent**: Baker–Wurgler investor sentiment change,
 - offering size, firm age, asset size, underwriter rank, VC backing, secondary share overhang, and past industry return.

2.6 Unobservables and timing

- The main unobservable is investor attention itself.
- Fundamental firm news may also affect search, so the paper must separate **attention** from **information**.
- Timing is crucial: the paper checks whether SVI leads other proxies and whether price effects eventually reverse.

3 Models and empirical specifications (all equations + interpretation)

General note. The paper is empirical rather than structural. The main objects are the construction of abnormal search volume, a VAR comparing attention proxies, panel regressions explaining attention, retail-trading regressions, Fama–MacBeth return regressions, and IPO return regressions.

3.1 Abnormal search volume

The core variable is

$$\text{ASVI}_t = \log(\text{SVI}_t) - \log\left(\text{Med}(\text{SVI}_{t-1}, \dots, \text{SVI}_{t-8})\right).$$

- SVI_t : current-week search volume.
- $\text{Med}(\text{SVI}_{t-1}, \dots, \text{SVI}_{t-8})$: normal recent attention level.

Interpretation. A high ASVI is an unusual increase in investor attention to a stock, scaled relative to that stock’s own past attention.

3.2 VAR for timing among attention measures

The paper compares four weekly attention variables in a vector autoregression:

$$\log(\text{SVI}), \quad \log(\text{Turnover}), \quad |\text{Abn Ret}|, \quad \log(1 + \text{Chunky News}).$$

Each stock-level VAR includes a constant and a time trend.

Interpretation. If lagged SVI predicts the other variables, then search measures attention earlier than returns, turnover, or news counts.

3.3 Panel regressions explaining abnormal attention

The paper estimates weekly panel regressions of the form

$$\text{ASVI}_{it} = \alpha_t + \beta X_{it} + \varepsilon_{it},$$

where α_t are week fixed effects and X_{it} includes size, abnormal returns, abnormal turnover, news variables, analyst coverage, advertising, and lagged news intensity.

Interpretation. These regressions ask whether abnormal search can be fully explained by the standard indirect proxies of attention. A low fit means SVI contains its own information.

3.4 Retail-trading regressions

At the monthly level, the paper estimates equations of the form

$$\Delta y_{it} = \alpha_t + \beta \Delta \log(\text{SVI}_{it}) + \Gamma Z_{it} + u_{it},$$

where Δy_{it} is the change in log retail orders or retail turnover and Z_{it} includes returns, absolute returns, market capitalization, news, and other controls.

Interpretation. A positive β means that when search rises, retail trading rises directly. This is the clearest evidence that SVI measures the attention of actual trading households.

3.5 Fama–MacBeth return regressions for Russell 3000 stocks

The price-pressure tests take the form

$$\text{DGTWAbRet}_{i,t+h} = \beta_h \text{ASVI}_{it} + \delta_h (\text{ASVI}_{it} \times \log \text{Size}_{it}) + \phi_h (\text{ASVI}_{it} \times \text{RetailShare}_{it}) + \Theta_h W_{it} + e_{i,t+h},$$

for horizons covering weeks 1 through 4 and weeks 5 through 52.

- The dependent variable is the DGTW characteristic-adjusted abnormal return.
- Controls include firm size, retail trading share, abnormal product-search volume, abnormal returns, advertising, analyst coverage, and news measures.

Interpretation.

- $\beta_h > 0$ at short horizons means attention predicts near-term outperformance.
- A negative long-horizon effect implies reversal and favors the **price-pressure** story over a pure **fundamentals** story.
- The interaction terms test whether attention matters more in small and retail-heavy stocks, as the Barber–Odean mechanism predicts.

3.6 IPO first-day return regressions

The IPO specification is

$$\text{FirstDayReturn}_i = \alpha + \beta \text{ASVI}_i + \gamma \text{Media}_i + \delta \text{PriceRevision}_i + \eta \text{DSENT}_i + \Psi X_i + \varepsilon_i,$$

where X_i contains standard IPO controls.

Interpretation. This asks whether pre-IPO retail attention predicts first-day IPO underpricing beyond the standard IPO variables.

3.7 IPO long-run return regressions

For long-run IPO performance, the paper estimates

$$\text{LongRunReturn}_i = \alpha + \beta_1 \text{ASVI}_i + \beta_2 \text{FirstDayReturn}_i + \beta_3 (\text{ASVI}_i \times \text{FirstDayReturn}_i) + \Lambda X_i + \nu_i.$$

The dependent variable is measured using raw returns or returns adjusted by industry or by a size/book-to-market matched portfolio from weeks 5 to 52 after issue.

Interpretation. The key object is the interaction term. If IPOs with high first-day returns only reverse when they also had high pre-IPO search, then those high first-day returns look like attention-driven price pressure.

3.8 Alternative information interpretation

The authors consider an alternative story in which search simply captures good fundamental information more quickly than standard variables do.

They address this by:

1. controlling for APSVI, abnormal product-search volume, and
2. testing for long-run reversal.

Interpretation. If the effect were only fundamental information, product-search volume should subsume SVI and there should be no later reversal. The data do not support that view.

4 Identification and instruments (what makes the estimates credible)

4.1 What fails in the standard approach

- Extreme returns and turnover are equilibrium outcomes, so they mix attention with many other forces.
- News coverage does not guarantee that investors actually read the news.
- In IPO settings, standard trading-based attention measures do not exist before the stock starts trading.

4.2 What identifies the new approach

- **Search is revealed attention.** Searching a stock ticker is a direct action that reflects active attention.
- **Ticker searches minimize ambiguity.** This makes SVI more tightly linked to financial attention than company-name searches.
- **Timing in the VAR.** SVI systematically leads the indirect attention proxies.
- **Direct retail-flow evidence.** Dash-5 regressions show that increases in SVI predict increases in retail orders and turnover.

4.3 Why the evidence points to retail attention

- Institutional investors usually have Reuters or Bloomberg and do not need Google in the same way retail investors do.
- SVI strongly predicts Dash-5 retail flow.
- The relation is much stronger at Madoff than at NYSE/Archipelago, consistent with SVI capturing less sophisticated retail investors in particular.
- Price pressure is stronger in smaller stocks and in retail-dominated stocks.

4.4 Why the price-pressure interpretation is credible

- The short-run price response is positive but later reverses.
- Product-search volume does not subsume the effect of stock search.
- The regressions include a broad battery of controls: returns, turnover, news, analysts, advertising, media coverage, price revision, sentiment, and standard IPO characteristics.
- SVI is the only stock-specific attention measure that predicts both an initial run-up and a later reversal.

4.5 Relation to the literature

- Provides a direct measure of investor attention for the limited-attention literature.
- Offers a direct test of the Barber–Odean hypothesis that retail investors buy attention-grabbing stocks.
- Extends the IPO literature by showing that search-based retail attention helps explain both first-day returns and later underperformance.

5 Tables and figures

5.1 Figure 1 and Table II: what SVI is and why it is different

Panel A. Google Trends search for “diet, cranberry”



Panel B. Google Trends search for “MSFT, AAPL”

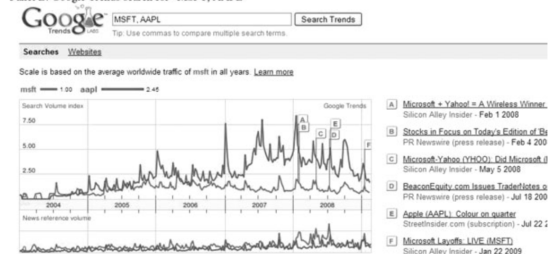


Figure 1. Illustrations of Google Trends search. Panel A represents the graphical output for a Google Trends search of “diet, cranberry.” The graph plots weekly aggregate search frequency (SVI) for both “diet” and “cranberry.” The SVI for “diet” is the weekly search volume for “diet” scaled by the average search volume of “diet,” while the SVI for “cranberry” is the weekly search volume for “cranberry” scaled by the average search volume of “diet.” Panel B represents the graphical output for a Google Trends search of the terms “MSFT, AAPL.” The graph plots weekly SVI for both “MSFT” and “AAPL.” The SVI for “MSFT” is the weekly search volume for “MSFT” scaled by the average search volume of “MSFT,” while the SVI for “AAPL” is the weekly search volume for “AAPL” scaled by the average search volume of “MSFT.”

Table II
Correlations

The table shows the correlations among variables of interest measured at weekly frequency. The variables are defined in Table I. The sample period is from January 2004 to June 2008.

	log(SVI)	log(Name.SVI)	Absolute Abn Ret	Abn Turnover	log(1 + News)	log(1 + ChunkyNews)	Frac.Neg.H4
log(Name.SVI)	0.093						
Absolute Abn Ret	0.059	0.093					
Abn Turnover	0.035	0.097	0.311				
log(1 + News)	0.050	0.155	0.199	0.181			
log(1 + ChunkyNews)	0.034	0.161	0.237	0.227	0.637		
Frac.Neg.H4	0.023	0.058	0.109	0.107	0.383	0.257	
Frac.Neg.LM	0.014	0.035	0.077	0.081	0.175	0.133	0.664

- **Read.** Figure 1 gives intuitive examples of search intensity. Search for “diet” and “cranberry” behaves like a direct public-attention measure. For stock tickers like MSFT and AAPL, search moves with attention but not identically with news coverage.
- **Table II.** Correlations between $\log(\text{SVI})$ and traditional proxies are all positive but low:
 - with Name SVI: about 0.093,
 - with absolute abnormal return: about 0.059,
 - with abnormal turnover: about 0.035,
 - with log news: about 0.050,
 - with chunky news: about 0.034.
- **Interpretation.** SVI is clearly related to attention, but it is not just a relabeling of returns, volume, or media coverage.

5.2 Table III and Table IV: SVI is timely and hard to explain away

Table III
Vector Autoregression (VAR) Model of Attention Measures

We compare four weekly measures of attention using vector autoregressions (VARs). The variables are defined in Table I. We run the VAR for each stock with at least 2 years of weekly data. We include both a constant and a time trend in the VAR. The VAR coefficients are then averaged across stocks and the associated p -values are reported below. These p -values are computed using a block bootstrap procedure under the null hypothesis that all VAR coefficients are zero. We start with the panel of residuals from the VAR and construct 10,000 bootstrapped panels. In the time-series dimension, we block-bootstrap with replacement using a block length of 23 weeks to preserve autocorrelation structure in the error terms. In the cross-sectional dimension, we also bootstrap with replacement. We repeat the VAR estimation in each bootstrapped panel, which allows us to build up the empirical distributions of the VAR coefficients. *, **, and *** represent significance at the 10%, 5%, and 1% level, respectively.

	Lagged 1 Week				R^2
	$\log(\text{SVI})$	$\log(\text{turnover})$	Absolute Abn Ret	$\log(1 + \text{Chunky News})$	
$\log(\text{SVI})$	0.5646*** (0.006)	−0.0022*** (0.01)	0.0489*** (0.01)	−0.0027*** (0.01)	56.47% 0.01
$\log(\text{turnover})$	0.0532** (0.005)	0.4467*** (0.01)	0.5197*** (0.01)	−0.0298*** (0.01)	38.82% 0.01
Absolute Abn Ret	0.0046*** (0.01)	0.0015*** (0.01)	0.0418*** (0.01)	−0.0011*** (0.01)	3.55% 0.06
$\log(1 + \text{Chunky News})$	0.0683** (0.02)	0.0270*** (0.01)	0.2071** (0.05)	0.0197*** (0.01)	3.19% 0.01

Table IV
Abnormal SVI (ASVI) and Alternative Measures of Attention

The dependent variable in each regression is abnormal ASVI. ASVI and independent variables are defined in Table I. Robust standard errors clustered by firm are in parentheses. *, **, and *** represent significance at the 10%, 5%, and 1% level, respectively. The sample period is from January 2004 to June 2008.

	(1)	(2)	(3)	(4)	(5)
Intercept	−0.099*** (0.006)	−0.096*** (0.006)	−0.095*** (0.007)	−0.096*** (0.007)	−0.096*** (0.007)
$\log(\text{Market Cap})$	0.001** (0.000)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.001** (0.000)
Absolute Abn Ret	0.131*** (0.012)	0.127*** (0.012)	0.127*** (0.012)	0.127*** (0.012)	0.129*** (0.012)
Abn Turnover	0.003*** (0.000)	0.003*** (0.000)	0.003*** (0.000)	0.003*** (0.000)	0.003*** (0.000)
News Dummy	0.001 (0.001)				
Chunky News Dummy		0.004*** (0.001)	0.004*** (0.001)	0.004*** (0.001)	0.004*** (0.001)
$\log(1 + \# \text{ of Analysts})$			0.000 (0.001)	0.000 (0.001)	0.000 (0.001)
Advertising Expense/Sales				0.007 (0.011)	0.010 (0.011)
$\log(\text{Chunky News Last Year})$					−0.001** (0.001)
Observations	411,930	411,930	411,930	411,930	411,930
Week fixed effects	YES	YES	YES	YES	YES
Clusters (firms)	2,435	2,435	2,435	2,435	2,435
R^2	0.03304	0.03315	0.03315	0.03315	0.03318

- **Table III.** In the VAR, lagged SVI positively predicts turnover, absolute abnormal returns, and chunky news. This is one of the strongest pieces of evidence that search captures attention earlier than conventional variables.
- **Table IV.** When ASVI is regressed on the usual attention proxies, the fit is tiny: the R^2 is only about 3.3%.
- **Interpretation.** Existing proxies explain only a small fraction of abnormal search activity, which means SVI contains distinct information.

5.3 Table V: SVI captures retail investor trading

- **Panel A.** A 1% increase in SVI predicts about

Table V
ASVI and Individual Trading Reported by Dash-5

We measure individual trading using orders (market and marketable limit) and trades contained in SEC Rule 11a-1.5 (Dash-5) reports. Panel A examines orders and trades reported by all market centers. We consider orders in two order size categories: (1) 100 to 1,999 shares and (2) 100 to 9,999 shares. Panel B considers orders in the 100 to 9,999 shares size category, examines different market centers separately (columns 1 through 4), and compares individual trading order/turnover response to concurrent SVI changes (columns 5 and 6) using a paired sample design. Madoff (columns 1 and 2) refers to Bernard L. Madoff Investment Securities LLC; NYSE/ARCH (columns 3 and 4) refer to the New York Stock Exchange (for NYSE-listed stocks) and Archipelago Holdings (for NASDAQ-listed stocks). In both panels, we regress monthly changes (log difference) in the number of individual orders (Δ Order) or monthly changes (log difference) in the individual turnover (Δ Turnover) on several variables. These include monthly SVI change (Δ SVI), alternative measures of attention and other stock characteristics. Δ SVI is the difference between the logarithm of SVI during month t and the logarithm of SVI during month $t - 1$, aggregated from weekly SVI. Other independent variables are defined in Table I. All regressions contain monthly fixed effects. Robust standard errors, reported in parentheses, are clustered at the stock level. *, **, and *** represent significance at the 10%, 5%, and 1% level, respectively. The sample period is from January 2004 to June 2008.

Panel A. Regressions of monthly Dash-5 reported order and turnover changes by order sizes				
	Order Size: 100–1,999 shares		Order Size: 100–9,999 shares	
	Δ Order (1)	Δ Turnover (2)	Δ Order (3)	Δ Turnover (4)
Δ SVI ($t-1, t$)	0.0925*** (0.0100)	0.0919** (0.00915)	0.103*** (0.0107)	0.131*** (0.0116)
Log/Market Cap ($t-1$)	-0.00670*** (0.000609)	-0.00754*** (0.000645)	-0.00713*** (0.000671)	-0.0108*** (0.000759)
Ret (t)	0.118*** (0.0209)	0.122*** (0.0241)	0.0989*** (0.0265)	0.0722 (0.0281)
Absolute Ret (t)	0.911*** (0.0480)	1.023*** (0.0460)	1.049*** (0.0500)	1.503*** (0.0540)
Chunky News Dummy (t)	0.0874*** (0.00360)	0.0942*** (0.00285)	0.0924*** (0.00310)	0.125*** (0.00286)
Advertising Expense/Sales ($t-1$)	-0.0429*** (0.0130)	-0.0346*** (0.00977)	-0.0506*** (0.0125)	-0.0506*** (0.0112)
Constant	0.139*** (0.0155)	0.145*** (0.0155)	0.156*** (0.0158)	0.179*** (0.0183)
Control Variables	YES	YES	YES	YES
Month fixed effect	YES	YES	YES	YES
Observations	108,954	108,954	108,954	108,954
Number of clusters (stock)	2,866	2,866	2,866	2,866
R^2	0.250	0.268	0.262	0.300

(continued)

Table V—Continued

Panel B. Regressions of monthly Dash-5 reported order and turnover changes by market center						
	Madoff		NYSE/ARCH		Comparison	
	Δ Order (1)	Δ Turnover (2)	Δ Order (3)	Δ Turnover (4)	Δ Order (5)	Δ Turnover (6)
Δ SVI ($t-1, t$)	0.264*** (0.0317)	0.297*** (0.0355)	0.090*** (0.0105)	0.104*** (0.0132)	0.166*** (0.0218)	0.204*** (0.0256)
Δ SVI + Madoff					0.139*** (0.0258)	0.0851*** (0.0274)
Madoff					0.000440 (0.00223)	0.0223*** (0.00233)
Log/Market Cap ($t-1$)	-0.0117*** (0.00202)	-0.0122*** (0.00207)	-0.00889*** (0.00094)	-0.0129*** (0.00713)	-0.00411*** (0.00132)	-0.00841*** (0.00132)
Ret (t)	0.154*** (0.0372)	0.0772 (0.0437)	0.0999*** (0.0173)	0.0647 (0.0199)	0.0418 (0.0284)	-0.0870*** (0.0331)
Absolute Ret (t)	1.299*** (0.0526)	1.570*** (0.0622)	1.901*** (0.0271)	1.418*** (0.0338)	1.344*** (0.0405)	1.622*** (0.0405)
Chunky News Dummy (t)	0.0668*** (0.00997)	0.0913*** (0.0121)	0.0936*** (0.00301)	0.120*** (0.00364)	0.0759*** (0.00678)	0.0901*** (0.00841)
Advertising Expense/Sales ($t-1$)	-0.104* (0.0630)	-0.0954 (0.0642)	0.00255 (0.00643)	-0.0328*** (0.00638)	-0.0713 (0.0610)	-0.0568 (0.0658)
Constant	0.255*** (0.0480)	0.251*** (0.0492)	0.175*** (0.0148)	0.229*** (0.0167)	0.0579* (0.0203)	0.119*** (0.0349)
Control variables	YES	YES	YES	YES	YES	YES
Month fixed effect	YES	YES	YES	YES	YES	YES
Observations	35,280	35,280	103,253	103,253	52,837	52,837
Number of Clusters (Stock)	1,358	1,358	2,743	2,743	862	862
R ²	0.131	0.127	0.299	0.291	0.173	0.191

Table VI
ASVI and Russell 3000 Stock Returns

This table reports the results from Fama-MacBeth (1973) cross-sectional regressions. The dependent variable is the DGTW abnormal return (in basis points) during the first 4 weeks and during weeks 5 to 52. Independent variables are defined in Table I. All variables are cross-sectionally demeaned (so the regression intercept is zero) and independent variables are also standardized (so the regression coefficients can be interpreted as the impact of a one-standard-deviation change). Standard errors are computed using the Newey–West (1987) formula with eight lags. *, **, and *** represent significance at the 10%, 5%, and 1% level, respectively. The sample period is from January 2004 to June 2008.

	Week 1 (1)	Week 2 (2)	Week 3 (3)	Week 4 (4)	Week 5–52 (5)
ASVI	18.742*** (7.000)	14.904** (7.561)	3.850 (6.284)	-1.608 (6.903)	-28.912 (17.162)
Log Market Cap $\times$ ASVI	-21.182*** (6.508)	-15.647** (6.768)	-4.710 (6.516)	4.290 (6.398)	16.834 (88.624)
Log Market Cap	2.653 (3.023)	3.858 (3.160)	3.144 (3.063)	3.575 (3.186)	-39.229 (67.405)
Percent Dash-5 Volume $\times$ ASVI	3.552** (1.639)	1.904 (1.522)	1.687 (1.612)	-2.744 (1.717)	16.258 (23.822)
Percent Dash-5 Volume	1.607 (1.644)	1.351 (1.652)	1.486 (1.659)	0.364 (1.711)	119.901*** (31.765)
APSVI	-2.532*** (0.930)	-1.379 (0.990)	-0.701 (0.808)	-0.704 (0.639)	2.286 (9.909)
Absolute Abn Ret	1.314 (1.879)	-2.389 (1.979)	-1.128 (1.563)	-0.463 (1.405)	-1.510 (28.505)
Advertising Expense/Sales	-4.012* (2.237)	-4.686** (2.228)	-3.959* (2.172)	-4.153* (2.234)	-162.210*** (52.414)
Log(1 + # of analysts)	-3.747** (1.548)	-4.547*** (1.741)	-3.961** (1.769)	-4.120*** (1.769)	-173.875*** (29.683)
Log(Chunky News Last Year)	-5.157 (3.370)	-5.549* (3.272)	-4.349 (3.292)	-5.409 (3.558)	-14.999 (80.730)
Chunky News Dummy	3.610* (2.025)	1.378 (2.424)	-3.825 (2.483)	-0.058 (1.910)	32.466 (28.441)
Abn Turnover	2.398** (1.204)	2.309** (1.144)	2.022 (1.404)	0.316 (1.098)	10.531 (10.109)
Observations per week	1,499	1,498	1,497	1,496	1,414
R^2	0.0142	0.0119	0.0112	0.0111	0.0170

- 0.0925% more retail orders and 0.0919% more retail turnover for the 100–1,999 share group,
- 0.103% more retail orders and 0.131% more retail turnover for the 100–9,999 share group.
- **Panel B.** The response is much stronger at Madoff:
 - order response: 0.264% at Madoff versus 0.092% at NYSE/Arch,
 - turnover response: 0.297% at Madoff versus 0.104% at NYSE/Arch.
- In the matched sample, the extra response at Madoff is about 0.109% for orders and 0.0951% for turnover.
- **Interpretation.** SVI is most strongly tied to less sophisticated retail attention.

5.4 Table VI: temporary price pressure in Russell 3000 stocks

- **Week 1.** A one-standard-deviation rise in ASVI predicts 18.742 bps of abnormal return.
- **Week 2.** It predicts another 14.904 bps.
- **Weeks 3 and 4.** The effect is small and fades.
- **Weeks 5–52.** The coefficient is -28.912 bps, indicating substantial long-run reversal.
- The interaction with size is negative, so the short-run effect is stronger for smaller firms.
- The interaction with Percent Dash-5 Volume is positive, so the short-run effect is stronger in retail-heavy stocks.
- **Interpretation.** This is the core evidence for the Barber–Odean mechanism.

5.5 Figure 2 and Figure 3: attention around IPOs

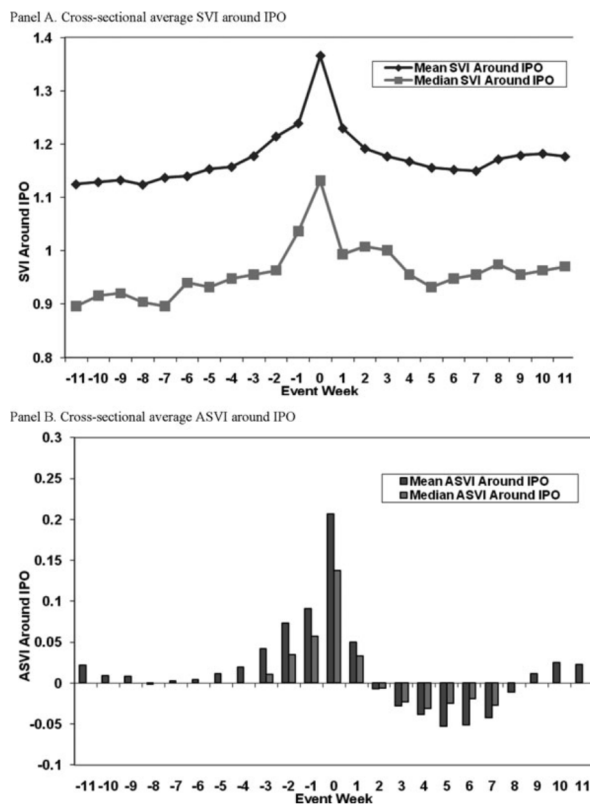


Figure 2. Average SVI and Abnormal SVI (ASVI) around IPO. Panel A plots the cross-sectional mean and median of the search volume index (SVI; in logarithm) around the week of the IPO. Panel B plots the cross-sectional mean and median of the ASVI around the week of the IPO. Week 0 is the week of the IPO. The sample period is from January 2004 to December 2007. There are 185 IPOs with valid SVI in this sample.

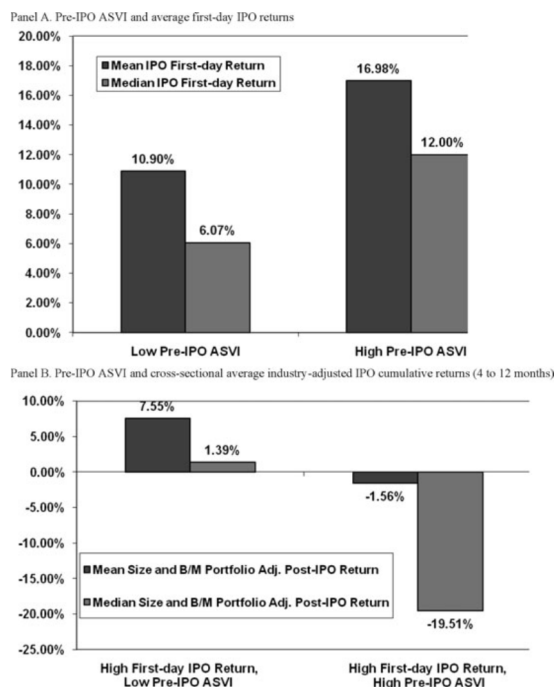


Figure 3. Pre-IPO ASVI, average first-day IPO returns and long-run IPO returns. Panel A plots pre-IPO ASVI and average first-day returns. Panel B plots pre-IPO ASVI and the size and book-to-market matched portfolio adjusted cumulative abnormal returns from week 5 to week 52. The sample period is from January 2004 to December 2007. There are 185 IPOs with a valid SVI in the sample.

- **Figure 2.** Search attention jumps around the IPO week. Mean SVI rises by about 20% during the IPO week and then largely reverts within 2 to 3 weeks.
- **Figure 3, Panel A.** IPOs with low pre-IPO ASVI have average first-day returns of about 10.90%, while IPOs with high pre-IPO ASVI have about 16.98%.

Table VIII
Pre-IPO Abnormal Search Volume (ASVI) and IPO First-Day Return

This table regresses IPO first-day returns on pre-IPO week abnormal search volume (ASVI) and IPO characteristics. The dependent variable is the individual IPO's first-day return. Independent variables are defined in Table I. The sample period of IPOs is from 2004 to 2007. Only regular and common stock IPOs (CRSP share class in 10 and 11) traded on NYSE, Amex, and NASDAQ with a valid SVI (searched using company names) are retained in the sample. Only IPOs with the first available CRSP closing price less than or equal to 5 days from the IPO date are retained. Standard errors (in parentheses) are clustered by the offering year and quarter. *, **, and *** represent significance at the 10%, 5%, and 1% level, respectively.

	Dependent Variable: IPO First-Day Return								
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
ASVI	0.275** (0.101)				0.203** (0.0795)				0.189** (0.0705)
Media		0.0292* (0.0149)				0.0255** (0.0114)			0.0246 (0.0144)
Price Revision			0.460*** (0.0806)				0.358*** (0.0989)		0.350*** (0.101)
DSENT				0.0134 (0.0119)				0.0194* (0.00933)	0.0221* (0.0104)
Log(Offering Size)					0.0805*** (0.0130)	0.0724*** (0.0128)	0.0344 (0.0219)	0.0855*** (0.0150)	0.0168 (0.0177)
Log(Age)					0.0187 (0.0167)	0.00995 (0.0149)	0.0131 (0.0165)	0.0131 (0.0164)	0.0121 (0.0113)
Log(Asset Size)					-0.0452** (0.00987)	-0.0446*** (0.00963)	-0.0239*** (0.00799)	-0.0453*** (0.00940)	-0.0197** (0.00692)
CM Underwriter Ranking					-0.00331 (0.00367)	-0.000222 (0.00319)	0.00670 (0.00453)	-0.000851 (0.00382)	0.00531 (0.00406)
VC Backing					0.0430 (0.0289)	0.0468 (0.0313)	0.0555* (0.0270)	0.0463 (0.0311)	0.0576* (0.0286)
Secondary Share Overhang					-0.0330 (0.0245)	-0.0332 (0.0203)	-0.0221 (0.0218)	-0.0308 (0.0222)	-0.0345 (0.0216)
Past Industry Return					0.199** (0.0994)	0.259*** (0.0744)	0.128 (0.102)	0.227*** (0.0765)	0.185** (0.0866)
Constant	0.114*** (0.0146)	0.0539 (0.0409)	0.143*** (0.0125)	0.135*** (0.0126)	-0.747*** (0.185)	-0.713*** (0.179)	-0.301 (0.271)	-0.811*** (0.209)	-0.180 (0.221)
Observations	185	185	185	185	185	185	185	185	185
R ²	0.052	0.037	0.235	-0.001	0.217	0.214	0.288	0.194	0.340

- **Figure 3, Panel B.** Among IPOs with high first-day returns, those with high pre-IPO search show clear long-run underperformance; those with low pre-IPO search do not.
- **Interpretation.** Search data are especially valuable around IPOs because they exist before trading starts.

5.6 Table VIII: pre-IPO ASVI predicts first-day returns

- Standalone, the coefficient on ASVI is 0.275, so a one-standard-deviation increase in ASVI (0.168) implies a 4.62% higher first-day return.
- With controls, the coefficient remains around 0.203.
- In the full specification, it remains 0.189, which still implies an incremental 3.18% higher first-day return.
- **Interpretation.** Even after controlling for Media, Price Revision, DSENT, and standard IPO characteristics, search-based retail attention still matters.

5.7 Table IX: long-run IPO reversal is attention driven

- The crucial result is the negative and significant interaction between ASVI and first-day return in long-run return regressions.
- For example, the interaction coefficient is about -2.501 for industry-adjusted long-run returns and about -2.169 for size/book-to-market-adjusted long-run returns.
- **Interpretation.** High first-day returns reverse later only when they were accompanied by unusual retail attention before the IPO. That is exactly the pattern predicted by temporary attention-driven price pressure.

Table IX
Pre-IPO Abnormal Search Volume (ASVI) and Post-IPO Performance

This table considers the cumulative IPO raw return (Panel A), cumulative IPO return adjusted by cumulative industry returns (Panel B), and cumulative IPO return adjusted by cumulative size and book-to-market equity matched portfolio (excluding stocks issued in the past 5 years) returns (Panel C) during the 4 to 12 months after the IPO. The dependent variable in Panel A is the individual IPO's cumulative return during the $t+5$, $w+52$ week window after the IPO, where w is the week the company went public. The dependent variable in Panel B is the individual IPO's cumulative return during the $t+5$, $w+52$ week window after the IPO adjusted by the corresponding industry matched portfolio returns during the same event window. The dependent variable in Panel C is the individual IPO's cumulative return during the $t+5$, $w+52$ week window after the IPO adjusted by the corresponding size and book-to-market equity matched portfolio (excluding recent IPO stocks in the past 5 years) returns during the same event window. To generate the size and book-to-market equity matched portfolio returns of non-IPOs, we match the first available market capitalization of the IPO with the immediate past June's NYSE market capitalization quintile break point, and then match the IPO's book-to-market equity ratio with the portfolio of stocks of the closest book-to-market equity quintile within the matched size quintile. The book value of the IPO is the first available book value of equity immediately after the IPO, and the market equity is the first available market capitalization of the IPO. The independent variables are defined in Table I. The sample period of IPOs is from 2004 to 2007. Only regular and common stock IPOs (CRSP share class in 10 and 11) traded on NYSE, Amex, and NASDAQ with a valid SVI (searched using company names) are retained in the sample. Only IPOs with the first available CRSP closing price less than or equal to 5 days from the IPO date are retained. Standard errors (in parentheses) are clustered by the offering year and quarter. *, **, and *** represent significance at the 10%, 5%, and 1% level, respectively.

Panel A: Pre-IPO abnormal search volume (ASVI) and IPO returns					
	Dependent Variable: IPO Return				
	(1)	(2)	(3)	(4)	(5)
ASVI	-0.176 (0.244)	0.409 (0.442)	-0.182 (0.239)	-0.187 (0.248)	0.548 (0.346)
ASVI × First-Day Return		-3.065** (1.060)			-3.330** (1.438)
Media	0.0523 (0.0421)	0.0413 (0.0451)	0.0611 (0.0437)	0.0565 (0.0427)	0.0521 (0.0417)
Media × First-Day Return		-0.0413 (0.0441)			-0.0851 (0.0792)
Price Revision	-0.0421 (0.181)	0.0396 (0.184)	-0.0392 (0.180)	-0.0422 (0.179)	-0.0420 (0.182)
Price Revision × First-Day Return		-0.434 (0.375)			-0.144 (0.520)
DSENT	-0.0646 (0.0606)	-0.0501 (0.0645)	-0.0621 (0.0622)	-0.0656 (0.0616)	-0.0664 (0.0701)
DSENT × First-Day Return		0.0154 (0.183)			0.132 (0.247)
Overhang	-0.110 (0.176)	0.173 (0.235)	0.0030 (0.135)	-0.0323 (0.228)	-0.117 (0.227)
Log(Offering Size)	0.0411 (0.130)	0.0382 (0.132)	0.0423 (0.135)	0.0438 (0.129)	0.0411 (0.130)

(continued)

Table IX—Continued

Panel A: Pre-IPO abnormal search volume (ASVI) and IPO returns					
	Dependent Variable: IPO Return				
	(1)	(2)	(3)	(4)	(5)
Log(Age)	-0.0184 (0.0667)	-0.0217 (0.0681)	-0.0146 (0.0635)	-0.0227 (0.0662)	-0.0184 (0.0672)
Log(Asset Size)	-0.0159 (0.0588)	-0.0184 (0.0593)	-0.0155 (0.0568)	-0.0143 (0.0580)	-0.0161 (0.0570)
CM Underwriter	0.0279 (0.0221)	0.0297 (0.0230)	0.0289 (0.0223)	0.0279 (0.0224)	0.0279 (0.0222)
Ranking	-0.170 (0.174)	-0.186 (0.171)	-0.170 (0.175)	-0.168 (0.170)	-0.170 (0.175)
VC Backing	-0.170 (0.174)	-0.186 (0.171)	-0.170 (0.175)	-0.168 (0.170)	-0.170 (0.175)
Secondary Share	-0.170 (0.174)	-0.186 (0.171)	-0.170 (0.175)	-0.168 (0.170)	-0.170 (0.175)
Overhang	-0.425 (0.297)	-0.379 (0.292)	-0.417 (0.294)	-0.425 (0.297)	-0.357 (0.297)
Past Industry Return	-0.399 (1.164)	-0.343 (1.156)	-0.448 (1.235)	-0.450 (1.154)	-0.397 (1.163)
Constant	185 (0.002)	185 (0.011)	185 (0.003)	185 (0.003)	185 (0.004)

Panel C: Pre-IPO abnormal search volume (ASVI) and book-to-market equity/size matched portfolio adjusted IPO returns

	Dependent Variable: Size and B/M Matched Portfolio Adjusted IPO Returns				
	(1)	(2)	(3)	(4)	(5)
ASVI	-0.226 (0.173)	0.252 (0.363)	-0.226 (0.173)	-0.219 (0.174)	-0.227 (0.174)
ASVI × First-Day Return		-2.189** (1.013)			-2.239 (1.244)
Media	0.0394 (0.0389)	0.0316 (0.0412)	0.0394 (0.0397)	0.0425 (0.0401)	0.0371 (0.0388)
Media × First-Day Return		0.0025 (0.0001)			0.0392 (0.0406)
Price Revision	-0.0180 (0.185)	0.0398 (0.199)	-0.0180 (0.180)	-0.0181 (0.185)	-0.0171 (0.205)
Price Revision × First-Day Return		-0.226 (0.420)			-0.217 (0.566)
DSENT	-0.0366 (0.0426)	-0.0264 (0.0453)	-0.0366 (0.0447)	-0.0374 (0.0434)	-0.0551 (0.0572)
DSENT × First-Day Return		0.156 (0.179)			0.211 (0.169)
First-Day return	-0.116 (0.155)	0.0849 (0.211)	-0.116 (0.170)	-0.0572 (0.255)	-0.190 (0.220)
Log(Offering Size)	0.0447 (0.108)	0.0427 (0.109)	0.0447 (0.109)	0.0467 (0.107)	0.0452 (0.109)
Log(Age)	-0.0386 (0.0386)	-0.0410 (0.0386)	-0.0386 (0.0419)	-0.0419 (0.0386)	-0.0397 (0.0397)

(continued)

Panel B: Pre-IPO abnormal search volume (ASVI) and industry matched portfolio adjusted IPO returns

	Dependent Variable: Industry Matched Portfolio Adjusted IPO Return				
	(1)	(2)	(3)	(4)	(5)
ASVI	-0.192 (0.155)	0.359 (0.307)	-0.188 (0.157)	-0.194 (0.155)	0.849 (0.349)
ASVI × First-Day Return		-2.501*** (0.834)			-2.456** (1.038)
Media	0.0176 (0.0328)	0.00681 (0.0340)	0.0107 (0.0347)	0.0207 (0.0335)	0.0152 (0.0325)
Media × First-Day Return		0.0321 (0.0509)			-0.0063 (0.0690)
Price Revision	-0.0607 (0.174)	0.00603 (0.185)	-0.0637 (0.173)	-0.0607 (0.174)	-0.0597 (0.175)
Price Revision × First-Day Return		-0.326 (0.365)			-0.198 (0.468)
DSENT	-0.0612 (0.0445)	-0.0494 (0.0473)	-0.0632 (0.0462)	-0.0620 (0.0456)	-0.0802 (0.0563)
DSENT × First-Day Return		0.150 (0.167)			0.176 (0.167)
First-Day Return	0.0143 (0.173)	0.245 (0.188)	-0.0738 (0.178)	0.0727 (0.223)	-0.0621 (0.206)
Log(Offering Size)	0.0609 (0.101)	0.0586 (0.103)	0.0600 (0.102)	0.0629 (0.100)	0.0609 (0.101)

(continued)

Table IX—Continued

Panel C: Pre-IPO abnormal search volume (ASVI) and book-to-market equity/size matched portfolio adjusted IPO returns					
	Dependent Variable: Size and B/M Matched Portfolio Adjusted IPO Returns				
	(1)	(2)	(3)	(4)	(5)
ASVI	-0.226 (0.173)	0.252 (0.363)	-0.226 (0.173)	-0.219 (0.174)	-0.227 (0.174)
ASVI × First-Day Return		-2.189** (1.013)			-2.239 (1.244)
Media	0.0394 (0.0389)	0.0316 (0.0412)	0.0394 (0.0397)	0.0425 (0.0401)	0.0371 (0.0388)
Media × First-Day Return		0.0025 (0.0001)			0.0392 (0.0406)
Price Revision	-0.0180 (0.185)	0.0398 (0.199)	-0.0180 (0.180)	-0.0181 (0.185)	-0.0171 (0.205)
Price Revision × First-Day Return		-0.226 (0.420)			-0.217 (0.566)
DSENT	-0.0366 (0.0426)	-0.0264 (0.0453)	-0.0366 (0.0447)	-0.0374 (0.0434)	-0.0551 (0.0572)
DSENT × First-Day Return		0.156 (0.179)			0.211 (0.169)
First-Day return	-0.116 (0.155)	0.0849 (0.211)	-0.116 (0.170)	-0.0572 (0.255)	-0.190 (0.220)
Log(Offering Size)	0.0447 (0.108)	0.0427 (0.109)	0.0447 (0.109)	0.0467 (0.107)	0.0452 (0.109)
Log(Age)	-0.0386 (0.0386)	-0.0410 (0.0386)	-0.0386 (0.0419)	-0.0419 (0.0386)	-0.0397 (0.0397)

(continued)

6 Short summary

- **Method.** Measure investor attention directly with Google search intensity and compare it to standard proxies.
- **Identification.** Show that search differs from other attention measures, leads them in time, and is strongly connected to retail order flow.
- **Application.** Russell 3000 stocks from 2004–2008 and U.S. IPOs from 2004–2007.
- **Main result.** Abnormal search volume predicts retail trading, short-run positive returns, and long-run reversal, especially where retail investors are important.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- The paper establishes that Google search data provide a direct and useful measure of investor attention.
- This measure captures the attention of retail investors more than institutions.
- Retail attention generates temporary price pressure in ordinary stocks and in IPOs.

7.2 Economic intuition

- Buying requires choice from a large set of stocks, while selling usually requires already owning the stock.
- Therefore, attention shocks create net retail buying pressure.
- Because the marginal buyers are not especially informed, the price effects are temporary rather than permanent.
- The strongest effects show up exactly where retail demand should matter most: small stocks, retail-heavy stocks, and IPOs.

7.3 Takeaway

This paper turns “attention” from a vague behavioral idea into a measurable object. Its main lesson is that search-based investor attention is economically powerful: it changes retail trading and temporarily moves prices, but those price movements largely reverse because they reflect demand pressure more than fundamentals.